MATH 330 Homework 1

Elementary Logic

Exercise 1. *Use a truth table to show that* $\neg(\neg P) \Leftrightarrow P$ *is a tautology.*

Exercise 2. *Determine whether each proposition is a tautology, a contradiction, or neither.*

1. $(P \Rightarrow Q) \Rightarrow Q$
2. $(P \wedge (P \Rightarrow Q)) \Rightarrow Q$
3. $(P \Rightarrow Q) \Leftrightarrow (Q \Rightarrow P)$
4. $((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$
5. $(P \Rightarrow Q) \wedge (P \Rightarrow \neg Q)$
6. $P \wedge (P \Rightarrow Q) \wedge (P \Rightarrow \neg Q)$
7. $(\neg Q \wedge (P \Rightarrow Q)) \Rightarrow \neg P$
8. $((P \vee Q) \wedge \neg Q) \Rightarrow P$
9. $((P \vee Q) \Rightarrow R) \Rightarrow (P \Rightarrow R)$

Exercise 3. *Determine whether each proposition is logically equivalent to* $P \Rightarrow Q$.

1. *The converse:* $Q \Rightarrow P$
2. *The contrapositive:* $\neg Q \Rightarrow \neg P$
3. *The inverse:* $\neg P \Rightarrow \neg Q$

Exercise 4. *Use truth tables to verify:*

1. *disjunction is commutative;*
2. *conjunction is commutative;*
3. *disjunction is associative; and*
4. *conjunction is associative.*

Exercise 5. *Show that* $P \wedge (Q \vee R)$ *is not equivalent to* $(P \wedge Q) \vee R$.

Exercise 6. *Use truth tables to verify both distributive laws:*

1. $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$;
2. $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$.

Exercise 7. *Determine whether each syllogism is valid.*

1. *Some dogs are beagles. Snoopy is a dog. Hence Snoopy is a beagle.*
2. *All cats are independent. Some pets are cats. Hence some pets are independent.*
3. *No pigs fly. Porky is a pig. Hence Porky does not fly.*
4. *No orcs are ogres. All ogres are green. Hence no orcs are green.*

Exercise 8. *Negate each statement.*

1. *Every cloud has a silver lining.*
2. *If it is Tuesday, then this is Belgium.*
3. *There is a light at the end of every tunnel.*

Exercise 9. *Use truth tables to verify De Morgan's laws:*

1. $\neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$;
2. $\neg(P \vee Q) \Leftrightarrow (\neg P \wedge \neg Q)$.

Exercise 10. *Use a truth table to show that* $\neg(P \Rightarrow Q) \Leftrightarrow (P \wedge \neg Q)$.